H · MERIDIAN

MISSION CONCEPT — SATELLITE PAYLOAD

STORE-AND-FORWARD CONSTELLATION — THE MESH IS THE GROUND SEGMENT

AF-H-101

REV 0 · 2026-11-12

VISION — EARNED, NOT SKIPPED

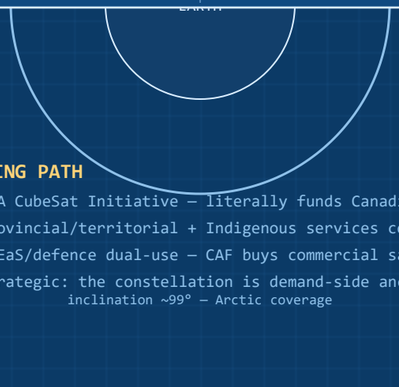
THE INSIGHT

- When the mesh needs backhaul beyond terrestrial reach, our own store-and-forward payload flies.
- Nobody else launching smallsats has a deployed rural ground network waiting — the mesh IS the satellite's user-terminal layer.
- A text message or sensor packet doesn't need Starlink bandwidth — it needs COVERAGE. A handful of CubeSats→smallsats can blanket Canada.
- Higher orbit is correct for THIS layer: bigger footprint per satellite = fewer needed. (Bandwidth wants LEO — that's a different game.)

ASCENT PATH

Stage	Vehicle	Cost class
[illegible]	[illegible] Canadian sat (GHGSat/MDA/univ)	[illegible]
[illegible]	[illegible] CubeSat — [illegible] partner	[illegible]
[illegible]	[illegible] (ISRO/JAXA/Arianespace/RL NZ)	[illegible]
[illegible]	[illegible] own ground ops	[illegible]
[illegible]	[illegible] coverage	[illegible]

ORBIT C



PAYLOAD — DERIVED FROM THE MESH

Element	Implementation
SDR	UHF/S-band store-and-forward transceiver — LoRa-derived waveform or CC1101-class
Protocol	Envelope-signed mesh-net format — satellite is just another hop
Storage	Onboard flash — holds envelopes until ground station passes
Bus	3U CubeSat (GOMspace/EnduroSat/CubeSat Shop) or smallsat
Ground segment	Shoulder mesh + community gateways = user terminals (already built)

FUNDING PATH

- CSA CubeSat Initiative — literally funds Canadian university smallsats.
- Provincial/territorial + Indigenous services contracts — northern connectivity mandate.
- IDEaS/defence dual-use — CAF buys commercial satcom; Canadian alternative is attractive.
- Strategic: the constellation is demand-side anchor for domestic launch (the summit).

MISSION PROFILE

- Coverage: polar LEO constellation blankets Canada incl. North — messaging/sensor tier, not broadband.
- Latency: store-and-forward hours, not seconds — right-sized for alerts, notices, telemetry.
- Capacity: each sat serves hundreds of nodes per pass — plenty for text/sensor traffic.
- Pathfinder: 2–6 sats prove the constellation, then scale to regional coverage.

HONEST STATUS

- Vision stage — no satellite exists. The mesh must prove itself first (L1→swarm→revenue).
- This sheet is the mission concept, not a spacecraft design — payload inherits mesh crypto/routing.
- Every earlier rung funds and justifies this one — the trailhead is the ground network.

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